

Course Structure

Machine Learning for Social Science

Course Policies

We will be using R extensively in this class. We will go over how to use R to do machine learning, but we will not focus on how to use R specifically.

If you are struggling with any of the material, **please let me know as soon as possible.**

Readings

Required Textbooks:

Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. New York, NY: Springer.

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). *An Introduction to Statistical Learning*. New York, NY: Springer.

Recommended Textbooks:

Foster, I., Ghani, R., Jarmin, R. S., Kreuter, F., and Lane, J. (Eds.). (2017). *Big Data and Social Science: A Practical Guide to Methods and Tools*. Boca Raton, FL: CRC Press Taylor & Francis Group.

Hastie, T., Tibshirani, R., Wainwright, M. (2015). *Statistical Learning with Sparsity: The Lasso and Generalizations*. Boca Raton, FL: Chapman & Hall/CRC.

Kuhn, M. and Johnson, K. (2013). *Applied Predictive Modeling*. New York, NY: Springer.

Canvas

All videos, assignments, and other course material will be available through **Canvas**.

All assignments, including the final paper will be submitted through **Canvas**.

Class Structure

We will be using a **flipped classroom structure**. In this course, you are responsible for **watching video recorded lectures and going through the readings**, and then attending class meetings where students have the chance to discuss the materials from a unit with the instructor. In general, the class time will be used for **discussion based on the forum posts, answer questions** as well as walking through **R code examples**.

Flipped Classroom Design

Before class:

- Watch the online video lectures.
- Do the readings.
- Post in the Discussion Forum for the week.

During class:

- Ask questions based on forum post discussion topics.
- Work through the R coding examples.

After class:

- Work on homework assignments.
- Set up an office hours appointment to clarify any remaining questions you might have.

Forum Posts

You must submit a **forum post every week** relating to the content covered in that week. You should post any questions you might have had about the content, including any R material that you may have struggled with. If you have no questions about the material, you can post comments about what you want to learn, or respond to another student.

We will use the questions to inform the first part of the discussion during class.

Class R Package

R tutorials for the week will be available through an R package:

<https://github.com/kimbrianj/mlforsocialscience>

You can install using:

```
devtools::install_github('kimbrianj/mlforsocialscience')
```


Credit points

Forum Posts: Weekly Forum Posts (starting next week)

- Response to the weekly video lectures and readings

Coursework: Assignments

- Short coding exercises to recap topics studied in class

ML project: Application of ML technique(s) in a social/ survey science context

- Term paper: Write-up of ML project
 - Includes motivation, methods, data, results, code (~ 5-10 pages)
 - Proposal in March.
 - Tentative due date: 4/29/2025

Credit points

Grade Distribution

- A missing submission will be scored as zero
- A submitted final paper is a precondition for passing the course
- Final grade: 50% assignments (averaged, without lowest score), 10% forum posts, 40% term paper

Homework for Today

Respond to the Week 1 Forum Posts with your research area of interest. What Machine Learning topics appeal to you the most? Which can you apply to your research?

Install and load the `mlforsocialscience` package.